energy-efficient. The design must also consider joint torque, arm geometry, cable management, and sensor placement for cameras, load cells, and distance sensors. Servos are suitable for smaller arms with limited rotational requirements, while stepper motors with encoders can be used where higher torque, greater angular range, and greater positioning accuracy are required.

For objects placed on a flat surface, a basic arm can operate with approximately four DoF (degrees of freedom): base rotation, shoulder, elbow, and gripper. This is sufficient when objects are considered within a two-dimensional X-Y workspace. For true three-dimensional manipulation, additional DoF and depth perception are required. Stereo or depth cameras can provide the object's Z-axis position, while additional joints allow the arm to orient and manoeuvre around obstacles.

For fully flexible manipulation in 3D space, six DoF are generally required: X, Y, and Z positioning, along with roll, pitch, and yaw orientation. Most industrial robotic arms therefore use six or more DoF, allowing the end-effector to reach and orient objects in different poses within the robot's workspace. Fig. 2 shows the completed arm setup, while Fig. 3 shows the arm assembly.

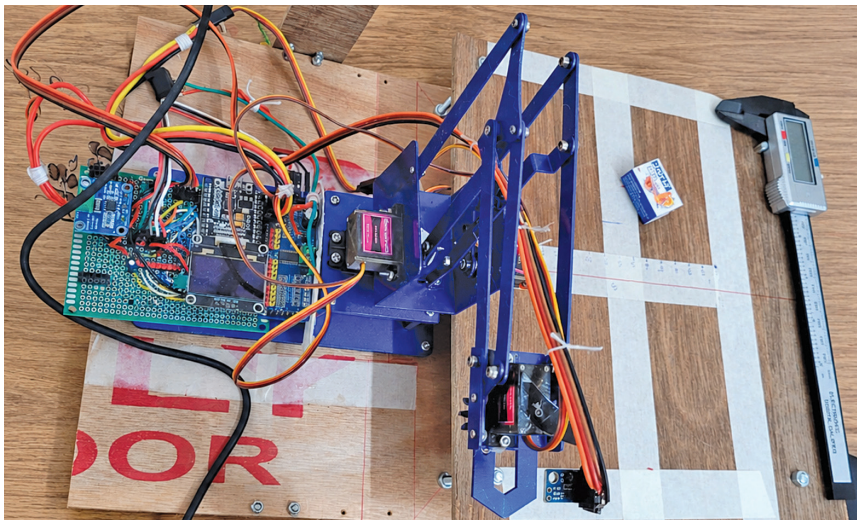


Fig. 2: Completed arm assembly

Circuit and control architecture

Fig. 4 shows the circuit diagram of the robotic arm. It is built around an ESP32-based NodeMCU-32S development board, which serves as

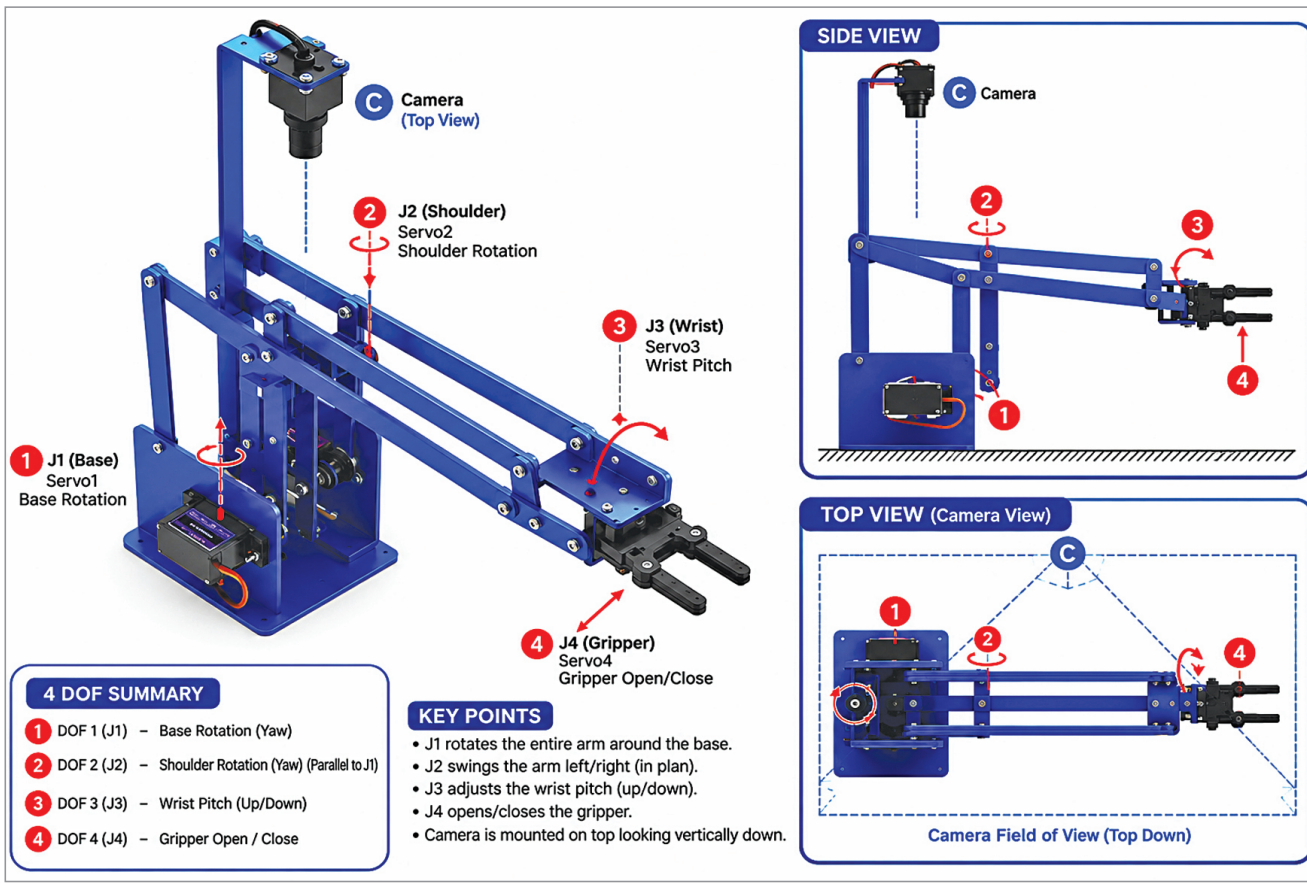


Fig. 3: Arm assembly